

Isogeometric Analysis — Practice 2

1D

In 2D ($n_{\text{sd}} = 2$), a curve ($n_{\text{pd}} = 1$) is defined by NURBS. Draw a curve $\mathbf{x}$,

$$\mathbf{t} = \frac{\mathbf{g}_1}{\|\mathbf{g}_1\|}, \quad (1)$$

and

$$\kappa \mathbf{n} = \|\mathbf{g}_1\|^{-1} \frac{d\mathbf{t}}{d\xi^1} \quad (2)$$

$$= \|\mathbf{g}_1\|^{-2} (\mathbf{I} - \mathbf{t}\mathbf{t}) \cdot \mathbf{x}_{,11}, \quad (3)$$

where $\mathbf{n}$ is a unit normal vector, $\mathbf{g}_1 = \frac{d\mathbf{x}}{d\xi^1}$, and $\mathbf{x}_{,11} = \frac{d^2\mathbf{x}}{d\xi^1 d\xi^1}$.

2D

From a p th order NURBS surface, write a program to calculate first and second fundamental forms at a give parametric position $\boldsymbol{\xi} = (\xi^1, \xi^2)$. The first fundamental form:

$$a_{\alpha\beta} = \mathbf{g}_\alpha \cdot \mathbf{g}_\beta, \quad (4)$$

and the second fundamental form:

$$b_{\alpha\beta} = \mathbf{x}_{,\alpha\beta} \cdot \mathbf{n}, \quad (5)$$

where

$$\mathbf{g}_\alpha = \frac{\partial \mathbf{x}}{\partial \xi^\alpha}, \quad (6)$$

$$\mathbf{n} = \frac{\mathbf{g}_1 \times \mathbf{g}_2}{\|\mathbf{g}_1 \times \mathbf{g}_2\|}, \quad (7)$$

$$\mathbf{x}_{,\alpha\beta} = \frac{\partial^2 \mathbf{x}}{\partial \xi^\alpha \partial \xi^\beta}. \quad (8)$$